

Benzel P7: reimplementation audit

Original-cell verification, unbounded identities, and dependency matching

Research audit prepared with ChatGPT

18 September 2026

1. Verdict and scope

The frozen turn-10 construction passes this audit. No counterexample, failed coefficient identity, invalid source-index certificate, or omitted canonical parameter was found. The residue-zero construction is complete at the level of the supplied written, exact-certificate-assisted argument. The full original Problem 7 assembly uses the explicitly matched generalized-compression theorem for the residue-two class.

This conclusion is based on a newly written original-cell checker, separately assembled symbolic expressions, reconstructed source constraints, exact arithmetic replays, and a fresh review of the universal mathematical deductions. It is not based on accepting the producer's recorded **PASS** values.

The checking implementations are independent of the producer's verification code. The research and this audit were nevertheless performed by the same assistant. This is **not external specialist acceptance, a Lean kernel certificate, or an independent audit of Sharma's manuscript**. The audit supplies inspectable evidence for a subsequent reviewer rather than silently changing a canonical publication status.

The frozen claim is the existence of original type-103 tilings of

$$W_h(d) = V(d + 3h, 2d + 3h), \quad d \geq 2, \quad 0 \leq h \leq t_d, \quad t_d = \frac{d(d-1)}{2}.$$

The claimed counts are $t_d - h$ right stones and $3h(h + d)$ bones. The supplied code additionally constructs the all-right-stone congruence class. Its full-P7 statement invokes an external theorem for the remaining congruence class; this audit retains that dependency.

The audited input is the **current/** subtree of the cumulative turn-10 archive, including its turn-7, turn-8, turn-9 and turn-10 mathematical dependencies. All **131 entries** of its source manifest matched, with no missing files and no hash mismatches. No mathematical source was repaired during this audit.

The input archive has SHA-256:

467427065b6d2612e105511b5350b6b6c15e7639dafda45e468c8670a3ae6e1f

The precise file identities are in **evidence/frozen-source-identity.json**. The candidate proof is a proof *package*: its 17-page PDF refers to full parent proofs, literal tile tables and rational certificates. The PDF alone is not its entire dependency closure.

2. What was rebuilt

The candidate constructors were treated as generators of objects to test, not as their own validity oracle. The new scripts live in `audit/`; unchanged candidate code and data live in `frozen/`.

Layer	New verification	Result
Original geometry	Reconstruct barycentric cells, allowed shapes and row intervals without the producer's geometry or coverage checker	PASS
Unbounded coefficient equations	Reassemble 18 identities in six formal variables using a separate symbolic implementation	PASS
Symbolic arithmetic	Replay the 18 exported identities again using only Python integer exponent dictionaries	PASS
Table semantics	Three formal equalities link source, strip and deletion parametrizations to their literal tables	PASS
Source ownership	Reconstruct 274 tile-owner identities and 6,559 exact affine certificates without importing the producer's constraint builder	PASS
Positive endpoint patches	Expand the bridge and all four exceptional patches as literal cell partitions	PASS
Integer kernel	Independently reconstruct matching-component relations and original-cell forest duals	PASS
Published input	Match the original cells, allowed tiles, parameters and positivity domain to compression Theorem 1.1	Exact dependency match

The algebra/source pipeline was run in normal and optimized Python. Both runs passed; their commands, source hashes and receipt hashes are recorded separately. The enlarged full-tiling sweep was run once. No claim is made that it was repeated in both modes.

3. Independent check on the original objects

3.1 Cells, tiles and an inverse coordinate map

The checker's original cell is a triple $(i, j, k) \in \mathbb{Z}^3$ with $i + j + k = 1$. It reads a candidate pair (x, y) through

$$(x, y) \mapsto (x, y, 1 - x - y),$$

whose inverse is projection onto the first two coordinates. These are Propp's barycentric cell coordinates [P].

For a candidate right stone, the new checker obtains the coordinatewise minimum p of the three triples and verifies

$$\sum p_i = 0, \quad \text{cells} = \{p + e_1, p + e_2, p + e_3\}.$$

It does not infer the orientation from a supplied label alone. For a bone, it computes the exact integer center and requires the three cells to be that center and its two adjacent translates in one of the original directions. The axial labels are checked against

$$H : (-1, 0, 1), \quad V : (0, 1, -1), \quad D : (1, -1, 0),$$

with both signs permitted along each centered bone. Floating-point coordinates, Boolean coordinates, repeated cells and wrong labels are rejected.

3.2 Region equality without using the producer's area

Write $L = 1 - a$, $U = b - 1$. For a fixed first coordinate i , the six original inequalities give the exact row

$$j_{\min} = \max \left(i + L, \left\lceil \frac{1 - i - U}{2} \right\rceil, L + 1 - 2i \right),$$

$$j_{\max} = \min \left(i + U, \left\lfloor \frac{1 - i - L}{2} \right\rfloor, U + 1 - 2i \right).$$

The first coordinate ranges from

$$\left\lceil \frac{1 + L - U}{3} \right\rceil \quad \text{to} \quad \left\lfloor \frac{1 + U - L}{3} \right\rfloor.$$

This follows from $i = (1 - (j - i) + (i - k))/3$. Rows with $j_{\min} > j_{\max}$ are empty.

For each row, the checker sorts *every cell occurrence* in the candidate and compares the sequence with every integer in this interval. It therefore detects holes and overlapping cells even when they cancel in a mere area count. It also tests the original inequalities on every cell directly.

Reflection is applied independently as the barycentric transposition $(i, j, k) \mapsto (i, k, j)$. It negates and reverses the three differences, carries $V(a, b)$ to $V(b, a)$, preserves right stones and exchanges the two appropriate bone labels. Its square is the identity.

3.3 Executed finite scope

The sweep checks every allowed h for $d = 2, \dots, 20$: **1,349 original parameter pairs** and their reflections. Larger tests cover each deletion residue at $k = 4, 5, 6, 9, 13, 20, 31$, two core sizes and the smallest and next exterior parameter; duplicates with the first sweep are removed. There are 64 additional original cases and 128 certificates including reflections. The all-stone class contributes 546 certificates for $2 \leq a \leq 40$, $a \leq b \leq 2a$, and their reflections.

In total:

Quantity	Executed count
Complete original/reflected tiling certificates	3,372
Placed tiles checked	59,403,188
Original cell occurrences checked	178,209,564
Deliberately invalid tiling certificates rejected	8

These are certificate counts, not a claim that all regions or reflected tilings are distinct. The finite sweep is implementation evidence; it does not establish an unbounded theorem by extrapolation.

4. The unbounded packing argument

The audit rechecked the retained turn-6 packing proof on its original cell carrier rather than replacing it by the finite sweep.

For the reflected first sector, put $\Delta = t_d - h$. Its prefix cells are

$$x = 1 - 2y - r + z, \quad 1 \leq y \leq h, \quad 0 \leq z \leq 3y - 1, \quad r = r(y + \Delta),$$

where $t_r < y + \Delta \leq t_{r+1}$. Since $y + \Delta \leq t_d$, one has $1 \leq r \leq d - 1$. The three original differences are

$$3y + r - 1 - z, \quad r - z, \quad 1 - 3y - 2r + 2z.$$

Their actual endpoint bounds establish containment in the reflected benzel. For a tail row $y = h + \ell$, the prefix is replaced by

$$x = \ell - 2h - d + 1 + z, \quad 0 \leq z < 3q_\ell, \quad q_\ell = \min(h, h + d - \ell).$$

Substitution gives

$$3h + d - 1 - z, \quad d - 3\ell - z, \quad 3\ell - 3h - 2d + 1 + 2z.$$

The inequalities $q_\ell \leq h$, $\ell + q_\ell \leq h + d$ give every original bound. Empty rows at $h = 0$ are retained as empty ranges.

Within a sector, different rows are disjoint. A putative overlap with a rotated sector has $1 \leq x < y$. In the prefix/prefix case, the two band inequalities force equal triangular-block index r and $y - x = r$; two values in one such block differ by at most $r - 1$. The tail/prefix case forces $d \leq y - x \leq d - 1$. The tail/tail case forces both $2x + y \leq 3h + d$ and $2x + y \geq 3h + d + 3$. The original rotations transfer these exclusions to every sector pair.

The rank map has a literal inverse. The three differences sum to zero and are pairwise unequal, because their cyclic differences are congruent to two modulo three. Rotate the cell so that its unique minimum is the first difference, $-r$. The cell is uniquely

$$p_{r,s} = (r - s, -s), \quad 0 \leq s < r,$$

and its rank is $t_r + s + 1$. Three rotations give exactly three cells per rank.

The first Δ ranks form Ω_Δ . To check band ownership, write $B = r - s$ and $c = 1 - 2r + 3s$. Of the three pre-reflection positions, the only possible owners give the complementary conditions

$$c \leq r(B + \Delta) \leq r, \quad \text{or} \quad c \geq 1 \text{ and } B + \Delta \leq t_c.$$

Together these are exactly $\Delta < t_r + s + 1$. The finite tail misses Ω_Δ ; its relevant difference is at least d , whereas a residual cell has differences at most $d - 1$. This checks the claimed stable complement, not just its cardinality.

The three phase domains give the original area independently as three copies of a triangle with three disjoint corner triangles removed. Per phase the count is

$$\binom{d + 3h}{2} - 2\binom{h}{2} - \binom{h + 1}{2}.$$

The resulting packing identity is

$$\partial\Gamma_\Delta(d, h) + \mathbf{1}_{\Omega_\Delta} = \mathbf{1}_{W_h(d)}.$$

For triangular $\Delta = t_m$, the rank inequalities give $\Omega_{t_m} = W_0(m)$. The explicit base stones then tile the entire residual.

5. Exact symbolic equations and their positive meaning

5.1 Separate coefficient engine

The new symbolic program imports no producer symbolic module. It reconstructs the expression from the frozen literal tables, the original offsets, and finite geometric sums. Both size parameters remain formal in

$$\mathbb{Z}[X^{\pm 1}, Y^{\pm 1}, M_x^{\pm 1}, M_y^{\pm 1}, K_x^{\pm 1}, K_y^{\pm 1}].$$

Specialization sends M_x, M_y, K_x, K_y respectively to X^m, Y^m, X^k, Y^k . Its kernel is generated by these four defining relations, with quotient inverse induced by the original cell variables.

For a nested family with monomial steps u, v , outer length I , and inner length $A + bi$, the exact identity is

$$\sum_{i=0}^{I-1} u^i \sum_{j=0}^{A+bi-1} v^j = \frac{\sum_{i=0}^{I-1} u^i - v^A \sum_{i=0}^{I-1} (uv^b)^i}{1 - v}.$$

Its derivation is the finite geometric identity in the inner sum. All actual lengths are checked to be nonnegative in the original domains; zero length is not replaced by a negative formal range. Direction-zero constant sums receive a separate literal treatment.

The affine cell rotation induces the linear map

$$\rho_* P(X, Y) = Y P(Y^{-1}, XY^{-1}),$$

and reflection induces $F_* P = Y P(XY^{-1}, Y^{-1})$. These are the cell-module maps, not unital ring homomorphisms. The underlying substitutions and the displayed translation monomial account for their exact effect on every tile boundary.

All denominator factors use only original cell directions and remain nonzero under parameter specialization. Clearing them takes place in an injective localization of an integer Laurent domain. Consequently a zero numerator gives the claimed original finite cell equation after specialization. The exact relation $1/(1 - z^{-1}) = -z/(1 - z)$ is used when choosing a common denominator; no parameter or tile coefficient is discarded.

5.2 What passed before specialization

The independent engine verifies 18 identities: the threefold family and its $k = 1$ endpoint; fixed $q = 1, 2, 4, 5$; and scaffold/endpoint pairs for the two uniform residues and all four exceptional cases. Every remaining numerator coefficient is zero. Three additional formal identities check that the parent source, long strip and corner-deletion tables agree with their written parametrizations.

The 18 complete rational expressions are exported, then replayed by a second small checker using only integer exponent dictionaries from the Python standard library. That replay also gives zero for every coefficient. This allows inspection of the entire arithmetic calculation without trusting SymPy's symbolic expression machinery. The geometric-sum interpretation and source-domain deductions remain the stated mathematical interpretation of those arithmetic receipts.

5.3 Positivity is an additional checked step

A signed identity alone would not prove a tiling. The audit separately reconstructs each negative generator's original owner and verifies its original index bounds, exclusion from the already removed source, and distinctness from the other negative generators.

There are **274 original affine tile-owner identities** and **6,559 exact nonnegative rational certificates**. The latter are equations between independently reconstructed integer affine vectors; each coefficient is checked with **Fraction**. For a desired bound, the receipt is an exact nonnegative combination of domain inequalities. For a forbidden overlap, it yields a negative constant from nonnegative constraints. A second standard-library checker replays all reconstructed linear receipts. No numerical optimization status is accepted.

The parent family lengths have 36 further bounds proved by nonnegative coefficients after the invertible domain map

$$k = v + 2, \quad m = u + 3v + 8, \quad v = k - 2 \geq 0, \quad u = m - 3k - 2 \geq 0.$$

The actual source incidence identity shows the initial scaffold is a positive packing. Distinct negative generators are present in that scaffold. Removing them once leaves a nonnegative integer vector; inserting the positive generators preserves nonnegativity. The final cell identity has coefficient one at each target cell and zero elsewhere, which proves containment, disjointness and full coverage.

6. The fragile endpoints and complete parameter coverage

Five local positive partition identities were re-expanded without the producer's tile or incidence functions: the eight-bone bridge and the four exceptional patches. Their common cell counts are respectively 27, 45, 54, 54 and 66. Both sides of every identity cover the same cells with coefficient one.

The zero-length first-residue channel at $m = 3k + 3$ still releases one old vertical bone. Its new-bone list is empty. That release is present in the formulas and certificates. The uniform beta-move range is empty, not negative, at $k = 4$. The $k = 2, 3$ cases have separate original constant patches; the uniform negative range is never used there.

The common source at the smallest exterior parameter $d = m$, $h = q$, maps to the finite tail using

$$j_{\text{tail}} = q - 1 - \ell, \quad j_{\text{tail}} = j_{\text{stable}} + q - y.$$

The inverse is $j_{\text{stable}} = j_{\text{tail}} + y - q$. Substitution into the original tail anchor gives the same three cells. At $d \geq m + 1$, $h \geq m + q$, so the identical generators lie in the original prefix. No smallest exterior case is omitted.

For $\Delta > 0$, choose the unique m with

$$t_{m-1} < \Delta \leq t_m, \quad q = t_m - \Delta.$$

Then $0 \leq q \leq m - 2$ and $m \leq d$. Conversely the equations recover Δ and $h = t_d - t_m + q$. For zero invariant use the explicit bone-only construction.

Canonical deletion case	Implemented construction
$q = 0$	Stable packing and triangular all-stone core
$q = 1, 2, 4, 5$	Four fixed-deletion tables
$q = 3k, k \geq 1$	Threefold family
$q = 3k + 1, k \geq 4$	First asymmetric family
$q = 3k + 2, k \geq 4$	Second asymmetric family
$q = 7, 8, 10, 11$	Four exceptional patches

Division of q by three exhausts this table. The only quotients below four in residues one and two give precisely the fixed and exceptional entries. All their core lower bounds are the canonical bound $m \geq q + 2$.

The two $d = 2$ cases have their own explicit partitions. This is a logical exhaustion proof, not the finite dispatch sweep.

As implementation controls, a separately written binary-search inverse was compared with the producer on 100,021 nonnegative integers, including triangular boundaries with core sizes up to 10^{100} . The dispatcher was checked at 100,001 deletion counts. Ten malformed or out-of-domain constructor inputs were rejected.

7. Integral comparison kernels and Split-Zero support

The audit independently reconstructs the graph attached to two actual disjoint bone matchings. The columns of $[\partial_-, -\partial_+]$ give, at each original cell, the difference of its old and new owners. A singly owned cell is attached to a ground vertex. Thus the signed incidence matrix is the transpose of an oriented graph incidence matrix after the explicit old/new sign convention; it is not merely compared by rank.

An equality between old and new boundaries forces equal coefficients along each shared-cell edge and zero at each grounded vertex. Its solutions are exactly constants on closed components, zero on all other components. This gives the primitive original integer generators of $B_- \cap B_+$.

Choose one new-bone pivot in each closed component. The quotient

$$K = B_+ / (B_- \cap B_+)$$

has basis the remaining new-bone classes. Its coordinate map subtracts the pivot coefficient on each closed component; its inverse puts zero at pivots. Each forest path supplies an integer original-cell dual. Direct evaluation on every old and new column verifies the annihilation and Kronecker identities.

The new implementation checked ten actual repair comparisons and two closed-component controls, producing **1,026 integer duals**. The closed controls include a single identical old/new bone and a three-horizontal-versus-three-vertical tiling of a 3×3 cell array; they exercise the primitive-relation quotient rather than only open graphs.

For the full support diagram, a nonbottom empty-release label still carries the original vacancy support. The scalar actions are

$$e(E, x) = (E, 0), \quad \tau(E, x) = (\perp, 0).$$

The incoming boundary and its supported-zero companion have the same original source and target labels. The fibre quotient descends against an arbitrary split-linear target because $f(E, y) + f(E, 0) = f(E, y)$; no cancellation assumption on that target is introduced.

The augmented differential and its actual transitions are

$$\widehat{d}_E(r, n) = \partial n - r v_E, \quad \widehat{j}_{EE'}(r, n) = (r, jn + r \eta_{EE'}).$$

The original identity $v_{E'} = j v_E + \partial \eta_{EE'}$ proves the cochain square, and addition of the disjoint released-bone sets proves composition. Charge-one nonnegative integral cycles identify exactly with original positive patch tilings. Their inclusion into integral cycles retains every coefficient.

The reconstructed matching quotient maps to the quotient allowing all contained bones. Its further kernel is $\text{im } \partial_{\text{all}} / (B_- + B_+)$. Neither that object nor the whole comparison kernel is replaced by the distinguished class. The cohomological deductions agree with the specified positive cell constructions; they are not used to manufacture a positive lift from a signed solution alone.

8. The full original problem and the published dependency

Propp's Problem 7 asks for type-103 tileability at nonnegative Conway–Lagarias invariant [P]. The audit checked the original statement and its cell convention rather than relying on the title of a later manuscript.

In the residue-zero class, after the explicit reflection if needed,

$$d = b - a, \quad h = (2a - b)/3$$

are integers. The original invariant in stone-cell units is $CL = 3(t_d - h)$. Admissibility gives $h \geq 0$; nonnegativity gives $h \leq t_d$. The values $d = 0, 1$ would force the invalid parameter pairs $(0, 0)$, $(1, 2)$, so every valid pair lands in the audited $d \geq 2$ construction.

8.1 Direct residue-one construction

For $a + b \equiv 1 \pmod{3}$, take any original cell c , and choose its unique candidate anchor $p = c - e_s$ whose difference residue is $2 - a \pmod{3}$. The three choices give all three residues; p has coordinate sum zero, so its three differences have that same residue.

The cell bounds put those anchor differences in $[-a, b]$. The least integer of the selected residue in that interval is $2 - a$, and the greatest is $b - 2$, because $a + b \equiv 1 \pmod{3}$. Hence every cell of $\{p + e_1, p + e_2, p + e_3\}$ lies in the original benzel. Every one of those three cells selects the same anchor. The map from cells to the unique anchor and offset therefore partitions the region into original right stones, with the inverse $(p, e_s) \mapsto p + e_s$. This establishes the full class directly.

8.2 Exact residue-two dependency match

Theorem 1.1 of Defant–Foster–Li–Propp–Young [C] applies to $2 \leq a \leq b \leq 2a$, $a + b \equiv 2 \pmod{3}$, and gives a count of tilings using right, rising and falling tiles. Its parameters in this report are

$$n = b + 1 - a \geq 1, \quad j = (2a - b - 1)/3 \geq 0,$$

with inverse $(a, b) = (n + 3j, 2n + 3j - 1)$. The actual count is

$$\prod_{\ell=1}^j \frac{(2\ell)!(2\ell + 2n - 2)!}{(\ell + n - 1)!(\ell + n + j - 1)!}.$$

Every factorial argument is nonnegative and every factor is positive; the empty product at $j = 0$ is one. The congruence gives $2a - b \equiv 1 \pmod{3}$, so $b \leq 2a$ really does imply $j \geq 0$. Thus the theorem provides existence throughout the required domain.

The geometric identification is explicit. With $\omega = e^{2\pi i/3}$, send the barycentric cell to $z = i + j\omega + k\omega^2$. For $z = x + iy$, the difference forms are

$$u = -x + y/\sqrt{3}, \quad v = -2y/\sqrt{3}, \quad w = x + y/\sqrt{3}.$$

The unit hexagon offsets $\pm 1, \pm\omega, \pm\omega^2$ change each form by at most one. Therefore the center bounds $[1 - a, b - 1]$ are exactly full-cell containment in the published enclosing hexagon. Its vertices are the stated $\omega^r(a\omega + b)$ and $-\omega^r(a + b\omega)$. The original bone directions become rising, vertical and falling directions. The inclusion of the published right/rising/falling tiles into the four permitted types preserves every original cell and commutes with incidence.

This is an exact theorem-interface audit. It is **not a fresh proof audit of every lemma of the compression paper**. Taking that published theorem as a cited input, the three congruence classes exhaust the full original P7 statement.

9. Findings, limits and provenance

No mathematical repair was required. The audit did not find a false universal identity, invalid released source, forgotten zero-length term, omitted residue, or invalid full tiling. The claimed residue-zero construction is supported by the rebuilt unbounded identities and source certificates together with the checked geometric and exhaustion arguments.

Two reporting limits must remain visible. First, the full-P7 result includes the published residue-two theorem as a dependency; the new constructor is not an independent implementation of generalized compression. Second, the unchanged prior manuscript claim and older algebra attribution are not displaced by a successful audit of this construction. No first-discovery conclusion follows from this report.

This audit did not validate the complete 90-page Sharma proof. Its method-specific claims, seed tables and layer recursion would require their own audit. The supplied parameter translation is consistent with the result checked here, but agreement on the theorem statement does not identify the two constructions.

The numerical search used during discovery is not part of the acceptance calculation. The current trust base consists of the audited finite-sum and geometry deductions, the original tables, exact Python integer and rational arithmetic, an independently used integer polynomial implementation, and the matched published theorem. The second arithmetic replay does not require a computer algebra system. A fresh copied-directory run also passed all seven algebra/source checks and 24 additional original/reflected certificate checks (269,470 placements). This was a smaller portability replay, not a second full sweep. A proof-assistant kernel has not checked the complete argument.

The development log includes one corrected bug in the *new audit harness*: a loop initially treated metadata in `families.json` as a tile-family list. The harness was restricted to the explicit source/corner/long/deleted collections and rerun. This did not change or repair a candidate source or a mathematical identity. The completed normal and optimized runs are recorded after the correction.

A suitable result label is **passed reimplementation audit; external review not completed**. No remote repository was changed in this review. The frozen earlier research statuses are preserved rather than rewritten.

10. Reproduction

From the unpacked audit directory, use Python 3.13 and SymPy 1.14.0 for the recorded environment. No internet connection or optimizer is needed for the mathematical replays.

```
python audit/run_algebra_checks.py
python audit/run_algebra_checks.py --optimized
python audit/original_cell_check.py --max-d 20
```

The first two commands reconstruct the expressions and constraints, then replay the exact arithmetic and local checks. The last command performs the large original-cell sweep. It took approximately 12.4 minutes in the audit environment; runtime and memory vary by machine.

To replay the exported arithmetic receipts using only the standard library:

```
python audit/check_polynomial_receipts.py
python audit/check_linear_receipts.py
```

Those commands verify the exported identities themselves. The reconstruction scripts and the written interpretation establish how the receipt vectors arise from the original tile tables.

The bundle includes the unchanged candidate `current/` source subtree, all new checkers, command ledgers, detailed receipts, an editable LaTeX report and a compiled PDF. Source files from external

literature and the supplied Sharma PDF are not redistributed.

References and evidence locators

[P] James Propp, *Trimer covers in the triangular grid: twenty mostly open problems*, Problem 7; original barycentric conventions and stone/bone figures. The linked author-hosted manuscript was inspected on 18 September 2026, including the pages defining the cell carrier and stating Problem 7. Source: <https://www.samuelhofkins.com/OPAC/files/proceedings/propp.pdf>.

[C] Colin Defant, Leigh Foster, Rupert Li, James Propp and Benjamin Young, *Tilings of Benzels via Generalized Compression*, Theorem 1.1, arXiv:2403.07663v1. The theorem statement, parameter domain and original geometric conventions were checked on 18 September 2026. The arXiv abstract record identifies the journal publication as SIAM Journal on Discrete Mathematics 39(1), 146–162 (2025), DOI 10.1137/24M1648247. The publisher DOI page was not accessible in this session; the mathematical statement was read in the author version. Source: <https://arxiv.org/html/2403.07663v1>.

[F] Frozen turn-10 package and its retained mathematical sources, identified by the source archive hash in Section 1. Exact paths, hashes, independent results and run modes are recorded in the accompanying JSON ledger. This report does not treat the frozen producer receipts as new verification.